Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/13/2004
PI Name:	Matkowsky, Bernard Ph.D.		
Project Title:	FILTRATION COMBUSTION FOR MICROGRAVITY APPLICATIONS: (1) SMOLDERING AND FIRE SAFETY (2) COMBUSTION SYNTHESIS OF ADVANCED MATERIALS		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60208-3107	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/04/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNC04GA18G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The task involves (1) proposing/developing theoretical models describing the fundamental mechanisms for the phenomena under consideration, (2) performing complementary analysis and numerical work on the proposed models and mechanisms, and (3) comparing the results of these analyses to experiments. The objective is to investigate combustion in porous media with applications to (1) smoldering and fire safety, and (2) combustion synthesis of advanced materials, also referred to as self-propagating high-temperature synthesis (SHS). Both smoldering and combustion synthesis of advanced materials (SHS) contribute to fundamental science, and also represent important applications for microgravity combustion science; in smoldering to fire safety, and in SHS to the determination of optimal synthesis conditions in both normal and microgravity environments. Combustion synthesis appears to compete favorably with conventional technology, by achieving shorter synthesis times, and at lower cost, by employing the		

	internal energy of the combustion reactions rather than the costly external energyof a furnace, employed in conventional technology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A paper "Effects of Gas-Solid Nonequilibrium in Filtration Combustion" appeared in Combustion Science and Technology 175, in 2003. The abstract of the paper reads: To determine the effects of gas-solid nonequilibrium on forced filtration combustion (FC) waves, a two-temperature model is employed, with separate temperature fields for the gas and solid particles. We consider heterogeneous (gas/solid) combustion in a porous sample with a prescribed gas influx at the inlet. If the combustion wave travels in the same direction (opposite to) the direction of the gas flux it is referred to as coflow (counterflow) FC. We determine the effects of gas-solid nonequilibrium on various aspects of forced FC waves. First, we consider coflow FC, in which case the gas infiltrating through the hot product region significantly enhances propagation of the wave. For a relatively small gas flux, the infiltrating gas delivers heat from the hot product layer to the combustion layer, thus increasing the combustion tempertaure and hence the reaction rate. Propagation of such waves is controlled by conduction of the heat released in the reaction, to the prehet zone. Conductively driven coflow FC waves have been extensively studied using one-temperature models, which assume a very large rate of interphase heat exchange between the solid and the gas, so that thermal equilibrium is rapidly attained. However, if the gas flux is sufficiently large, the phases do not have sufficient time to equilibrate and hence one-temperature models are no longer valid. For relatively large gas flux, using the two-temperature model, we show that an alternative mechanism of enhancement occurs whereby the combustion temperature increases as a result of increasing the effective initial temperature ahead of the reaction zone. Propagation of such waves is controlled by convection of heat stored in the product, to the preheat zone. Convectively driven coflow FC waves depend on a pronounced temperature difference between the phases and cannot be described by a one-temperature model. Employing the two-tempertaure model we describe both conductively and convectively driven FC waves. We compare and contrast the two wave structures and their combustion characteristics, including propagation velocity, extinction limit, and the ability of the wave to accumulate energy. We also discuss the transition between the two types of waves. Next, we consider the effects of nonequilibrium on counterflow FC waves, in which the solid near the reaction site loses heat to the cool infiltrating gas. Counterflow waves are necessarily conductively driven. Extinction can occur if the gas flux is sufficiently large. We find that thermal nonequilibrium has a significant effect on the extinction limit and on the combustion characteristics.
Bibliography Type:	Description: (Last Updated: 11/08/2006)
Articles in Peer-reviewed Journals	Wahle, C.W., Matkowsky, B.J. "Effects of Gas-Solid Nonequilibrium in Filtration Combustion" Vol. 175, pp. 1389-1499 , Oct-2003